

Applying system thinking to Singapore's air transport system

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Introduction

Four main parts of Singapore's air transport system:

- ☐ Physical facilities
- ☐ Operation processes
- ☐ Management structures
- ☐ External influences



Key Challenge

**System Composition
&
Interface Complexity**

**Tension between Capacity
Expansion
&
Environmental Sustainability**

**Coordination
between Labour Constraints
&
Automation Transition**

Literature Review

System Structure & System Levels

Airport systems consist of multiple interconnected layers including terminals, airside operations, ground transport and resource management.

(Wu & Mengersen, 2013; Li et al., 2025)

System Integration & Interface Complexity

Airport performance is strongly affected by process coordination, interface management, and alignment across operational subsystems.

(Wu & Mengersen, 2013)

Expansion vs. Sustainability

Recent studies discuss aviation growth, decarbonisation pressure, technological change, and policy arrangements within the same framework.

(Raihan,2025)

Workforce Constraints & Automation

Labour shortages, digitalisation, job redesign, and skill adaptation have become closely connected issues in airport operations.

(Wandelt & Wang, 2024; Williady et al., 2024)

System Structure& Key Characteristics

3 Key Characteristics:

Increasing
complexity

Strong
interdependency

Multiple
trade-offs



System structure

Physical
Infrastructure

Operations
&
Services

Governance

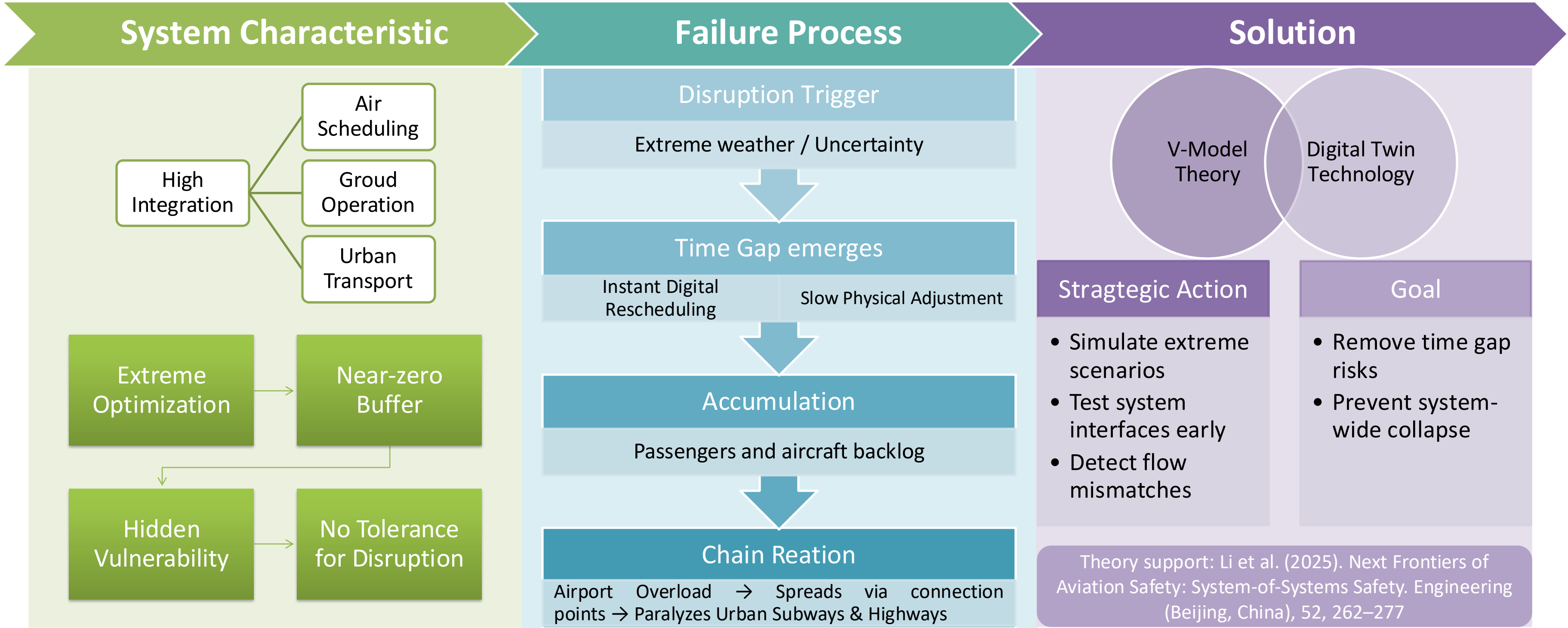
External
Environment



Core Challenge 1

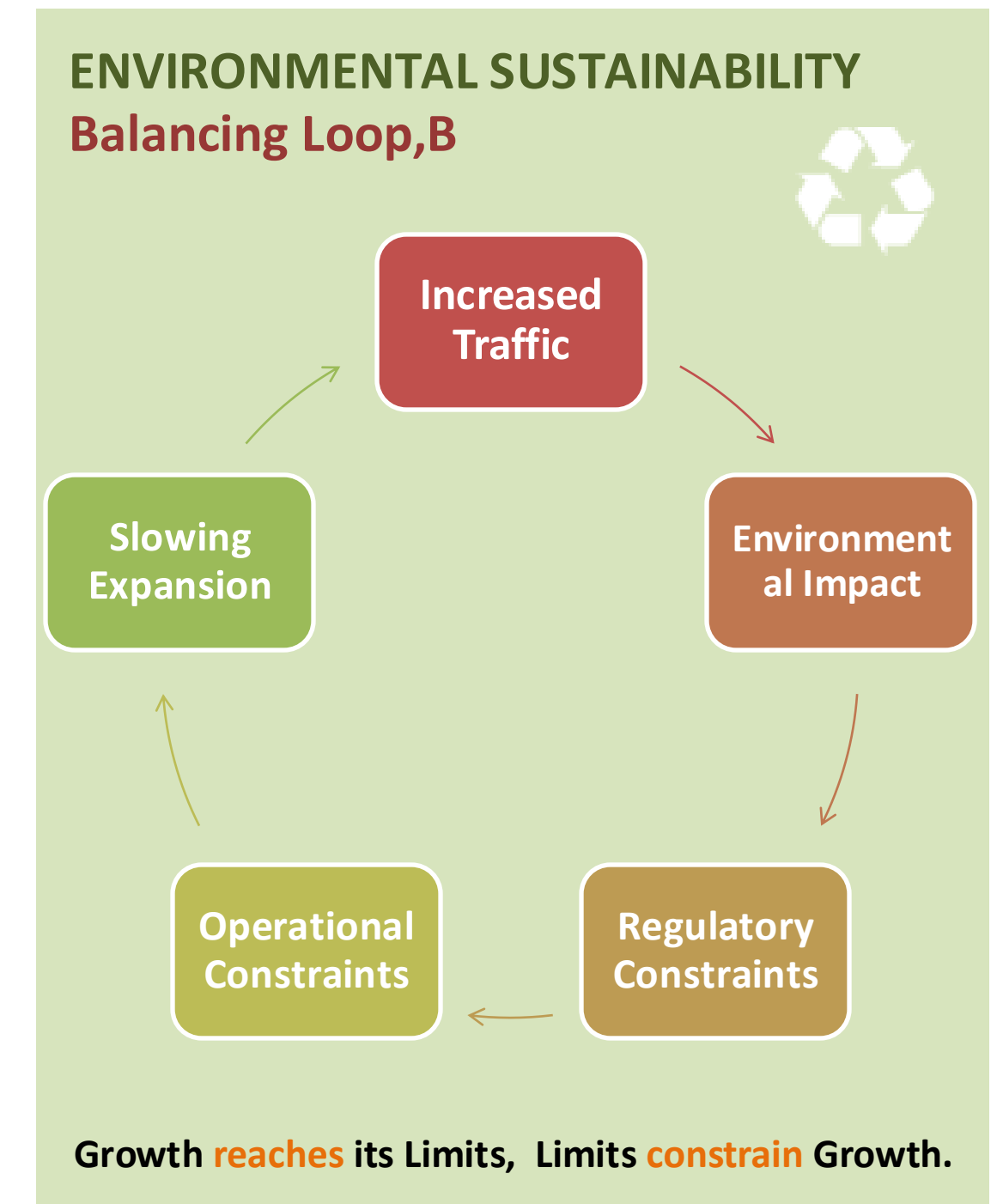
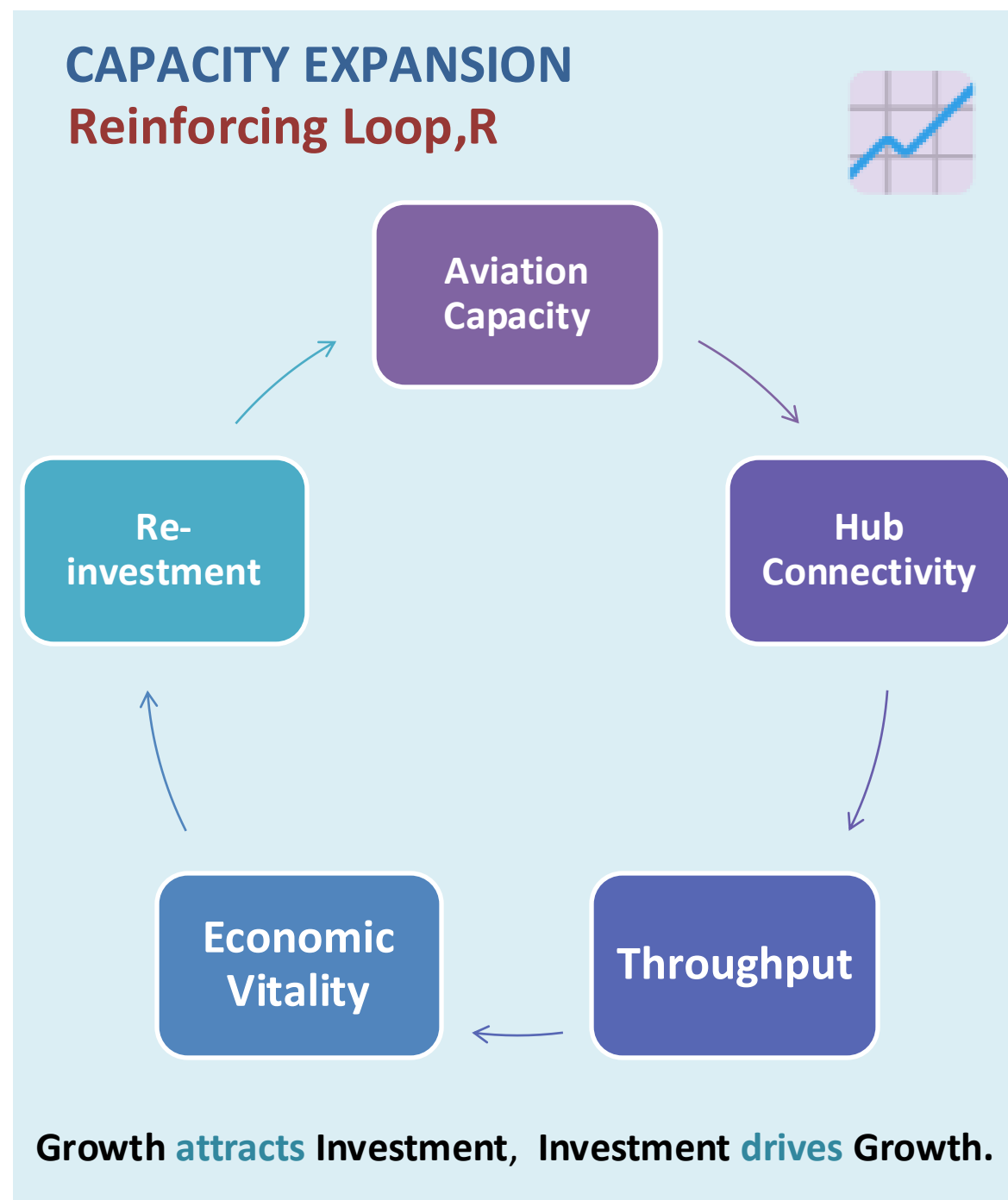
System Composition and Interface Complexity

High efficiency creates zero-buffer vulnerability, leading to cascading failure under disruption



Core Challenge 2

Tension between Capacity Expansion & Environmental Sustainability



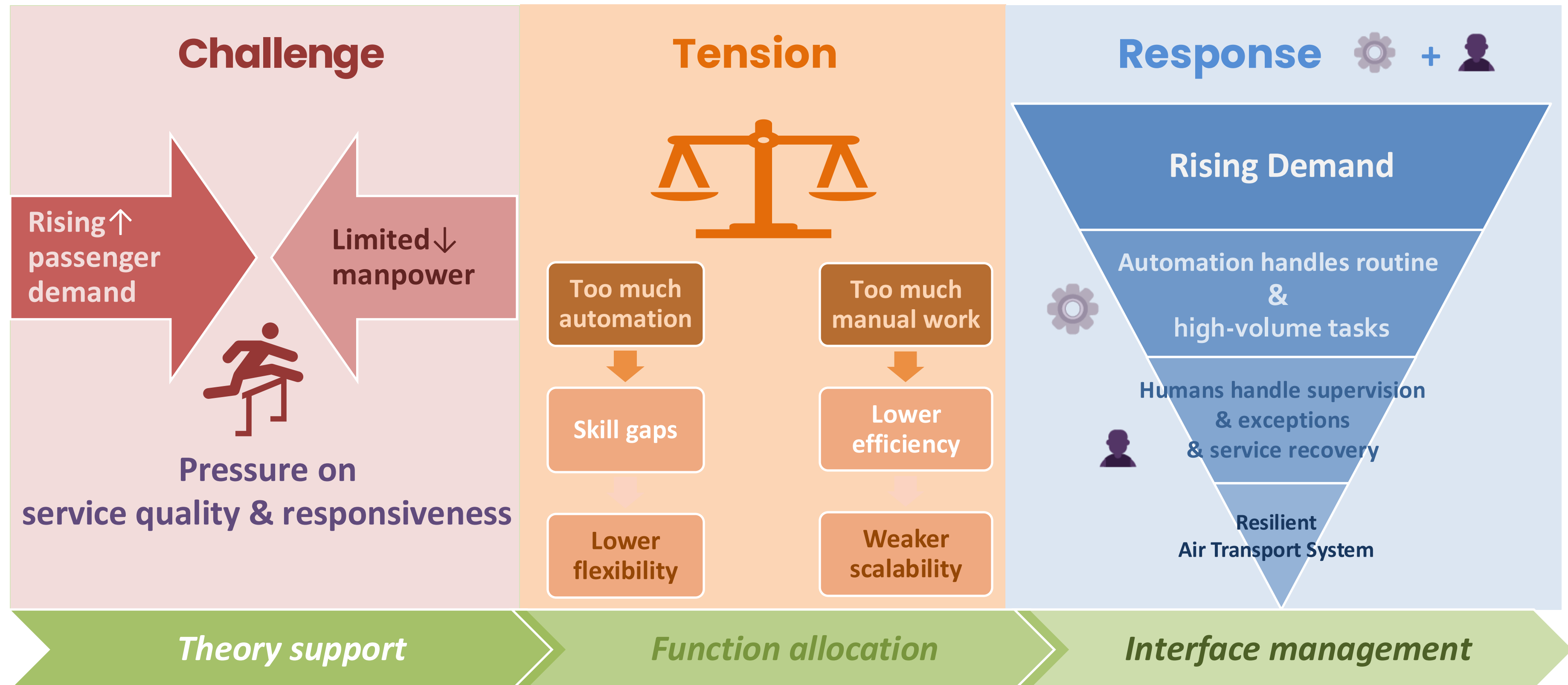
System Boundary Redefinition

Value-Based Prioritization

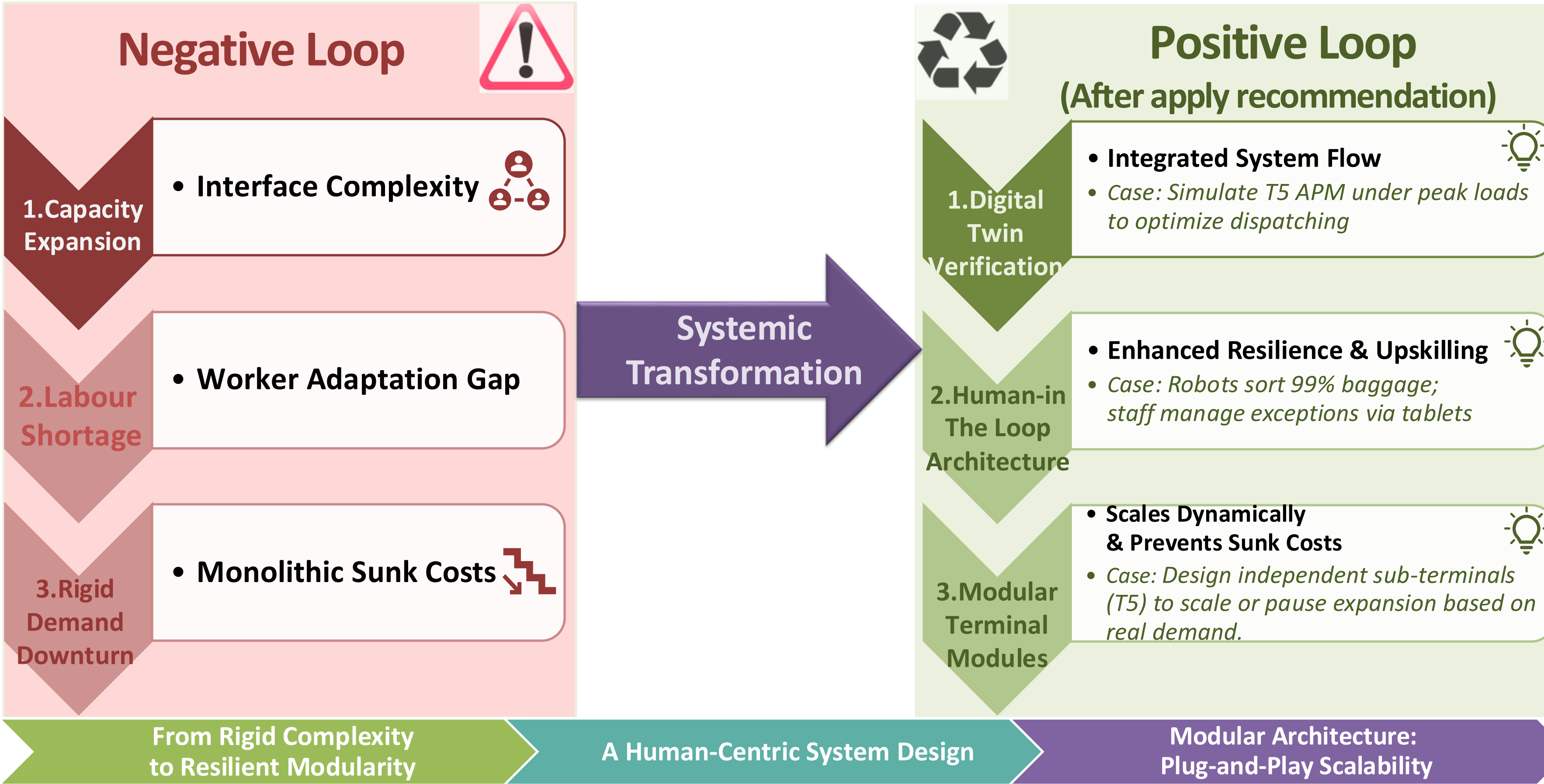
SAF Infrastructure Integration

Core Challenge 3

Coordination between Labour Constraints and Automation Transition



Key Recommendations: Transforming the Aviation System



Critical Reflection

A system engineering mindset

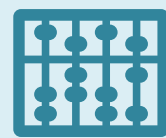


A Shift In Perspective



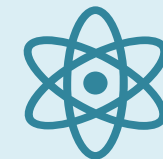
- To: "System Of Systems"
- From: Fragmented Issues (e.g., delays, labor shortages)

Use Of System Thinking



Complexity

- V-Model & Digital Twin for complex interfaces



Flexibility

- "Human-in-the-Loop"
- Modular Architecture



Balancing

- System Dynamics

Key Takeaway



Holistic Balancing > Local Optimization



Systems thinking is more than a tool — it is a mindset



Find root causes, design for a sustainable future

Thank You

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